Lecture 33

Section 6.4

The Gram-Schmidt Process

Goal: Produce an orthogonal basis from any basis.

Let's see what happens in $\mathbb{R}^2$.

Example: Suppose that $\mathcal{B} = \{\mathbf{x}_1, \mathbf{x}_2\}$ is a basis for W. We want to turn $\mathcal{B}$ into an orthogonal basis $\{\mathbf{v}_1, \mathbf{v}_2\}$ for W. So, we need $\text{span}\{\mathbf{v}_1, \mathbf{v}_2\} = W = \text{span}\{\mathbf{x}_1, \mathbf{x}_2\}$ and $\mathbf{v}_1 \bullet \mathbf{v}_2 = 0$. Recall that an orthogonal set of nonzero vectors is linearly independent.

Consider the projection of $\mathbf{x}_2$ onto $\mathbf{x}_1$, namely $\hat{\mathbf{x}_2} = \text{proj}_{\mathbf{x}_1}\mathbf{x}_2$. By the orthogonal decomposition theorem,

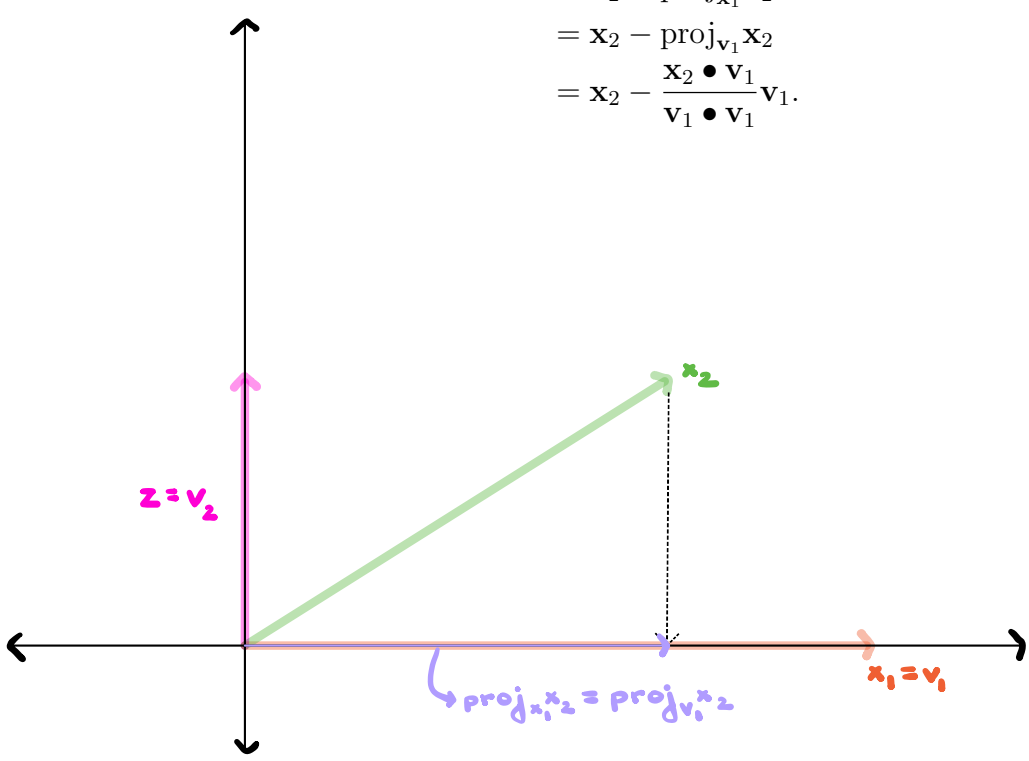
$$\mathbf{x}_2 = \hat{\mathbf{x}_2} + \mathbf{z}$$

where $\hat{\mathbf{x}_2}$ is a scalar multiple of $\mathbf{x}_1$ and $\mathbf{z}$ is orthogonal to $\mathbf{x}_1$. So, $\mathbf{x}_2 - \hat{\mathbf{x}_2} = \mathbf{z}$ is orthogonal to $\mathbf{x}_1$, i.e. $\mathbf{z} \bullet \mathbf{x}_1 = 0$. Therefore, $\mathbf{z}$ and $\mathbf{x}_1$ are linearly independent. Also, $\mathbf{z} \in \text{span}\{\mathbf{x}_1, \mathbf{x}_2\} = W$. So, we define

$$\mathbf{v}_1 = \mathbf{x}_1$$

and

$$\begin{aligned}
\mathbf{v}_2 &= \mathbf{z} \\
&= \mathbf{x}_2 - \hat{\mathbf{x}_2} \\
&= \mathbf{x}_2 - \text{proj}_{\mathbf{x}_1}\mathbf{x}_2 \\
&= \mathbf{x}_2 - \text{proj}_{\mathbf{v}_1}\mathbf{x}_2 \\
&= \mathbf{x}_2 - \frac{\mathbf{x}_2 \bullet \mathbf{v}_1}{\mathbf{v}_1 \bullet \mathbf{v}_1}\mathbf{v}_1.
\end{aligned}$$



The Gram-Schmidt Process: Suppose we have a basis $\{\mathbf{x}_1, \ldots, \mathbf{x}_m\}$ for a nonzero subspace W of $\mathbb{R}^n$. Let

$$\begin{aligned}
\mathbf{v}_1 &= \mathbf{x}_1, \\
\mathbf{v}_2 &= \mathbf{x}_2 - \frac{\mathbf{x}_2 \bullet \mathbf{v}_1}{\mathbf{v}_1 \bullet \mathbf{v}_1}\mathbf{v}_1, \\
\mathbf{v}_3 &= \mathbf{x}_3 - \frac{\mathbf{x}_3 \bullet \mathbf{v}_1}{\mathbf{v}_1 \bullet \mathbf{v}_1}\mathbf{v}_1 - \frac{\mathbf{x}_3 \bullet \mathbf{v}_2}{\mathbf{v}_2 \bullet \mathbf{v}_2}\mathbf{v}_2, \\
&\vdots \\
\mathbf{v}_m &= \mathbf{x}_m - \frac{\mathbf{x}_m \bullet \mathbf{v}_1}{\mathbf{v}_1 \bullet \mathbf{v}_1}\mathbf{v}_1 - \frac{\mathbf{x}_m \bullet \mathbf{v}_2}{\mathbf{v}_2 \bullet \mathbf{v}_2}\mathbf{v}_2 - \cdots - \frac{\mathbf{x}_m \bullet \mathbf{v}_{m-1}}{\mathbf{v}_{m-1} \bullet \mathbf{v}_{m-1}}\mathbf{v}_{m-1}.
\end{aligned}$$

Then, $\{\mathbf{v}_1, \ldots, \mathbf{v}_m\}$ is an orthogonal basis for W.

Note: Recall that you can create an orthonormal basis from an orthogonal basis by normalizing each vector.

Example: Suppose $W = \text{span}\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\}$ where

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \qquad \mathbf{x}_2 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \qquad \mathbf{x}_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

First, find an orthogonal basis for W. Then, find an orthonormal basis for W.

Solution: Let $\mathbf{v}_1 = \mathbf{x}_1$. Then,

$$\mathbf{v}_2 = \mathbf{x}_2 - \frac{\mathbf{x}_2 \bullet \mathbf{v}_1}{\mathbf{v}_1 \bullet \mathbf{v}_1}\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} - \frac{1+2+0}{1+1+1}\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

and

$$\begin{aligned}
\mathbf{v}_3 &= \mathbf{x}_3 - \frac{\mathbf{x}_3 \bullet \mathbf{v}_1}{\mathbf{v}_1 \bullet \mathbf{v}_1}\mathbf{v}_1 - \frac{\mathbf{x}_3 \bullet \mathbf{v}_2}{\mathbf{v}_2 \bullet \mathbf{v}_2}\mathbf{v}_2 \\
&= \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} - \frac{1+0+0}{1+1+1}\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \frac{0+0+0}{0+1+1}\begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \\
&= \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} - \frac{1}{3}\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \\
&= \begin{bmatrix} 2/3 \\ -1/3 \\ -1/3 \end{bmatrix}.
\end{aligned}$$

Note that we can multiply $\mathbf{v}_3$ by 3 to make it "nicer." So, consider

$$\mathbf{v}_3' = \begin{bmatrix} 2 \\ -1 \\ -1 \end{bmatrix}.$$

Therefore, $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3'\}$ is an orthogonal basis for W. To make this an orthonormal basis, we just need to divide each vector by its length. Note that

$$||\mathbf{v}_1|| = \sqrt{\mathbf{v}_1 \bullet \mathbf{v}_1} = \sqrt{1+1+1} = \sqrt{3},$$
$$||\mathbf{v}_2|| = \sqrt{\mathbf{v}_2 \bullet \mathbf{v}_2} = \sqrt{0+1+1} = \sqrt{2},$$
$$||\mathbf{v}_3'|| = \sqrt{\mathbf{v}_3' \bullet \mathbf{v}_3'} = \sqrt{\underbrace{\cancel{\frac{4}{9} + \frac{1}{9} + \frac{1}{9}}}_{4+1+1}} = \sqrt{\underbrace{\cancel{\frac{6}{9}}}_{6}} = \underbrace{\cancel{\frac{\sqrt{6}}{3}}}_{=\sqrt{6}}.$$

So, an orthonormal basis for W is

$$\left\{ \begin{bmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix}, \begin{bmatrix} 0 \\ \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}, \begin{bmatrix} \frac{2}{\sqrt{6}} \\ -\frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{6}} \end{bmatrix} \right\}.$$

QR Factorization of Matrices

The QR Factorization: Suppose A is an $m \times n$ matrix with linearly independent columns. Then, we can write $A = QR$, where Q is an $m \times n$ matrix whose columns form an orthonormal basis for Col A and R is an $n \times n$ upper triangular invertible matrix with positive entries on its diagonal.

Note: Recall that a matrix Q has orthonormal columns if and only if $Q^{-1} = Q^T$.

Example: Suppose

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$

Find a QR factorization for A.

Solution: In the previous example, we found an orthonormal basis of Col A. So, we use this basis to construct Q, namely

$$Q = \begin{bmatrix} \frac{1}{\sqrt{3}} & 0 & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} \end{bmatrix}.$$

By the above theorem, we have that $Q^{-1}A = R$, i.e. $Q^T A = R$. So,

$$R = Q^T A = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} \frac{3}{\sqrt{3}} & \frac{3}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{2}} & 0 \\ 0 & 0 & \frac{2}{\sqrt{6}} \end{bmatrix} = \begin{bmatrix} \sqrt{3} & \sqrt{3} & \frac{1}{\sqrt{3}} \\ 0 & \sqrt{2} & 0 \\ 0 & 0 & \frac{2}{\sqrt{6}} \end{bmatrix}.$$

Notice that the diagonal entries of R are the lengths of the vectors from the original orthogonal basis we computed using Gram-Schmidt.